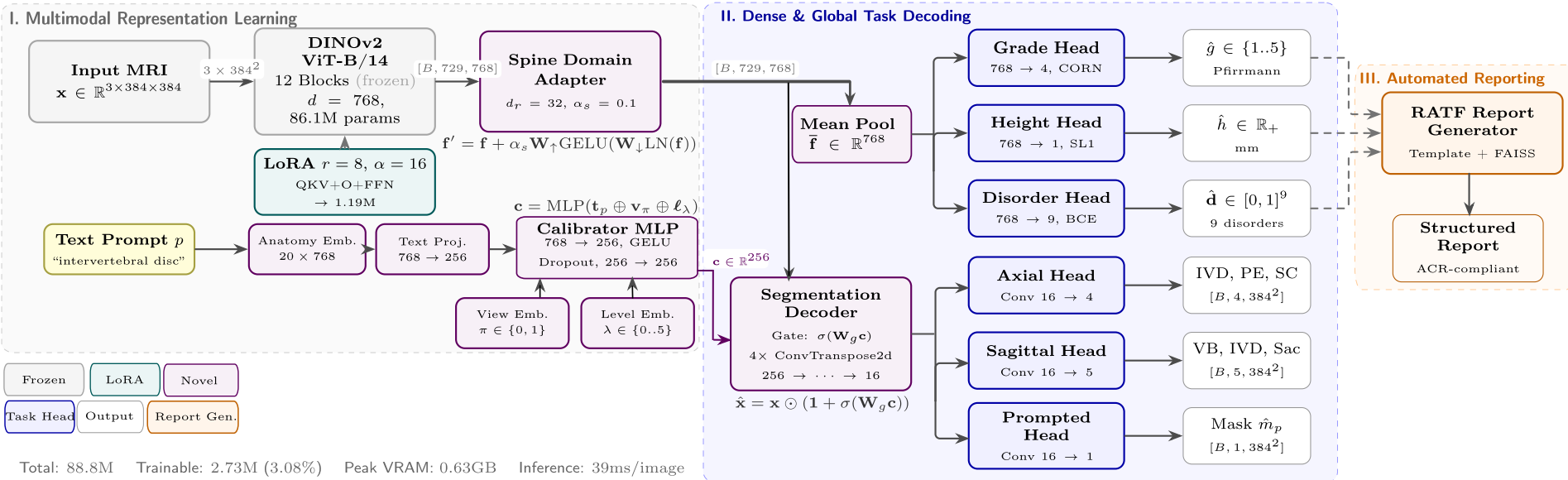

I. Multimodal Representation Learning
Input MRI
$\mathbf{x} \in \mathbb{R}^{3\times384\times384}$
3×384^2
DINOv2 ViT-B/14
12 Blocks (frozen)
$d = 768$,
86.1M params
$[B, 729, 768]$
Spine Domain Adapter
$d_r = 32$, $\alpha_s = 0.1$
$\mathbf{f}' = \mathbf{f} + \alpha_s \mathbf{W}_\uparrow \mathrm{GELU}(\mathbf{W}_\downarrow \mathrm{LN}(\mathbf{f}))$
LoRA $r = 8$, $\alpha = 16$
QKV+O+FFN
→ 1.19M
Text Prompt p
"intervertebral disc"
Anatomy Emb.
20 × 768
Text Proj.
768 → 256
$\mathbf{c} = \mathrm{MLP}(\mathbf{t}_p \oplus \mathbf{v}_\pi \oplus \boldsymbol{\ell}_\lambda)$
Calibrator MLP
768 → 256, GELU
Dropout, 256 → 256
View Emb.
$\pi \in \{0, 1\}$
Level Emb.
$\lambda \in \{0..5\}$
$\mathbf{c} \in \mathbb{R}^{256}$
II. Dense & Global Task Decoding
$[B, 729, 768]$
Mean Pool
$\bar{\mathbf{f}} \in \mathbb{R}^{768}$
Grade Head
768 → 4, CORN
$\hat{g} \in \{1..5\}$
Pfirrmann
Height Head
768 → 1, SL1
$\hat{h} \in \mathbb{R}_+$
mm
Disorder Head
768 → 9, BCE
$\hat{\mathbf{d}} \in [0, 1]^9$
9 disorders
Segmentation Decoder
Gate: $\sigma(\mathbf{W}_g \mathbf{c})$
4× ConvTranspose2d
256 → ··· → 16
$\hat{\mathbf{x}} = \mathbf{x} \odot (\mathbf{1} + \sigma(\mathbf{W}_g \mathbf{c}))$
Axial Head
Conv 16 → 4
IVD, PE, SC
$[B, 4, 384^2]$
Sagittal Head
Conv 16 → 5
VB, IVD, Sac
$[B, 5, 384^2]$
Prompted Head
Conv 16 → 1
Mask $\hat{m}_p$
$[B, 1, 384^2]$
III. Automated Reporting
RATF Report Generator
Template + FAISS
Structured Report
ACR-compliant
Frozen
LoRA
Novel
Task Head
Output
Report Gen.
Total: 88.8M
Trainable: 2.73M (3.08%)
Peak VRAM: 0.63GB
Inference: 39ms/image